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Gastroprotective and antioxidant potentials of ethanolic stem bark extract of *Margaritaria discoidea* (Euphorbiaceae) in rats

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ABSTRACT

Ethnopharmacological relevance: Decoctions prepared from the bark of *Margaritaria discoidea* (Baill.) G. L. Webster (Euphorbiaceae) are used in Nigeria and other parts of West Africa in the treatment of wounds and ulcers. The study was conducted to investigate the gastric ulcer protective effect of ethanolic stem bark extract of *M. discoidea* in rats.

Materials and methods: Antiulcer assays were performed using ethanol, indomethacin and pylorus ligation-induced ulcer models at the dose of 50, 100 and 150 mg/kg, p.o. The antioxidant effect of the extract was evaluated *in vitro* and by studying its effect on antioxidant enzymes (superoxide dismutase, catalase, and reduced glutathione) and lipid peroxidation in the stomach tissue of rats in ethanol-induced model. Solvent fractions (hexane, dichloromethane, ethyl acetate, butanol and aqueous) from the crude extract were investigated for antiulcerogenic effects in ethanol-induced ulcer model at the dose of 150 mg/kg. GC–MS analysis of the active hexane fraction was also carried out.

Results: The extract significantly ($P < 0.05$) reduced gastric lesion in ethanol and indomethacin-induced ulcer models. The extract had significant influence on *in vivo* antioxidant status in ethanol-induced model. In pylorus ligation-induced model, only the dose of 150 mg/kg showed significant reduction (88.89%, $P < 0.05$) of ulcer lesions. There was no significant reduction in the gastric juice volume and total acidity. The solvent fractions showed ulcer inhibition in varying degrees but significance ($P < 0.01$) was only observed in the hexane fraction. Ethyl esters of palmitic and linoleic acids were found as the major compounds in the GC–MS analysis of the hexane fraction.

Conclusion: Our results suggest that *M. discoidea* possesses gastroprotective activity possibly mediated through antioxidant mechanism. The data obtained in this study provide some support to the traditional use of *M. discoidea* in the treatment of gastric ulcer.

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1. Introduction

Gastric ulcer is the most common form of peptic ulcer and the most predominant of the gastrointestinal diseases (Falcao et al., 2008; Lakshmi et al., 2010). It is a chronic and recurrent disease, with multi-etiopathological factors. Stress, smoking, *Helicobacter pylori* infection and ingestion of non-steroidal anti-inflammatory drugs (NSAID) augment the gastric ulcer incidences (Vonkeman et al., 2007). Free radicals, particularly reactive oxygen species (ROS) have also been implicated in the mechanism of acute and chronic ulceration in the gastric mucosa (Sathish et al., 2011). An approach to manage gastric ulcer disease, therefore, is through the scavenging of ROS and the stimulation of the endogenous antioxidant enzymes in the stomach, in addition to the other approaches such as, the inhibition of gastric H^+K^+-ATPase and the elimination of *H. pylori* using antibiotics (Nartey et al., 2012).

The current trend of research is the investigation of medicines of plant origin because medicinal plants enjoy wide acceptability by the population and serve as cheaper alternatives to orthodox medicine (Akah and Nwabie, 1994), especially in the developing countries. In line with this, the potential gastroprotective and antioxidant properties of *M. discoidea* were investigated.

M. discoidea (Baill.) G. L. Webster (Euphorbiaceae) is a tree which can grow up to 30 m tall depending on its location. The stem is usually straight with rough, flaking bark which is greyish–brown on top and reddish beneath. It is widely distributed in Africa region where it is being used for treatment of various ailments. In Nigeria and Ghana, the decoction of the bark is used in the treatment of wounds and ulcers while in Malawi, the

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powdered bark extract is applied to swellings and inflammation for quick relief (Irvine, 1961; Burkill, 1994).

Previous pharmacological studies of *M. discoidea* have shown that the plant displays acaricidal (Kaaya et al., 1995), anti-inflammatory and analgesic (Adedapo et al., 2009), filaricidal (Cho-Ngwa et al., 2010) and cytotoxic (Johnson-Ajinwo et al., 2015) activities. Securinega alkaloids such as phyllochrysine and securinine (Mensah et al., 1988; Weenen et al., 1990; Fehler, 2000), betulinic acid (Calixto et al., 1998), hydroxylgenkwanin-8-C-[α-rhamno-pyranosyl-(1→6)]-β-glucopyranoside (margadiscoside), genkwanin-6-C-[α-rhamnopyranosyl-(1→6)]-β-glucopyranoside, kaempferol-3-O-α-rhamnopyranosyl-(1→2)-β-glucopyranoside-7-O-α-rhamnopyranoside and kaempferol-3-O-α-rhamnopyranosyl-(1→2)-[α-rhamno-pyranosyl-(1→6)]-β-glucopyranoside-7-O-α-rhamnopyranoside (Ekuadzi et al., 2014) have been reported isolated from the plant. No report was found in the literature that demonstrates the gastroprotective property of the plant. Therefore, the present study was carried out to evaluate the gastroprotective potential of the ethanolic extract of *M. discoidea.*

2. Materials and methods

2.1. Plant material and extract preparation

The stem bark of *M. discoidea* was collected in Ikire (7.35 latitude, 4.18 longitude), Osun state in south-western Nigeria in the month of February, 2013. The specimen was authenticated by Mr. T. K. Odewo at the Herbarium of the Department of Botany, Faculty of Science, University of Lagos, with a voucher specimen number, LUH 5552.

The stem bark was cleaned, cut into small pieces and dried in the oven at 40 °C. The dried material was powdered using a laboratory mechanical grinder. The powdered material (900 g) was macerated twice with absolute ethanol (2.5 L) for 48 h, at room temperature. The extract was filtered and evaporated to dryness in a water bath at 40 °C. The extract obtained was dark brown in color and the percentage yield was 5.98% (w/w).

2.2. Fractionation of the crude extract

The extract (30 g) was dissolved in 80 ml of water and fractionated by successive solvent extraction with n-hexane (2 ml × 200 ml), dichloromethane (2 ml × 750 ml), ethyl acetate (2 ml × 750 ml) and n-butanol saturated with water (2 ml × 750 ml) in a separating funnel. Each extract as well as remaining aqueous phase after solvent extractions was evaporated to dryness to yield hexane (HEX, 0.33 g), dichloromethane (DCM, 7.10 g), ethylacetate (EtOAc, 1.92 g), butanol (BuOH, 1.92 g) and aqueous (AQU, 3.23 g) extracts, respectively.

2.3. Phytochemical screening

Preliminary phytochemical screening of the crude extract was carried out using established procedures (Harborne, 1998; Trease and Evans, 2002). Quantification of polyphenolic classes including the estimation of total phenolics (Wolfe and Liu, 2003), flavonoids (Ordonez et al., 2006) and proanthocyanidins (Sun et al., 1998) was carried out.

2.4. GC–MS analysis of the active hexane fraction

Agilent 6890N GC system furnished with an auto sampler (Agilent 7683 injector series) was coupled to a 5973 Network mass selective detector (GC–MS) (based on a quadruple mass separator) was used to run the hexane fraction of the plant. A J&W Scientific HP-5MS silica fused capillary column (30 m × 0.25 mm i.d. × 0.25 μm film thickness) was used with helium as the carrier gas at a constant flow rate of 1.0 ml/min. Splitless injection of 2 μl of the sample was automatically done by an injector (injector 7683 series) on the instrument from a syringe 10 μl. The oven temperature range was set at 70 °C and ramped at 4 °C/min to 250 °C. The injector temperature was set at 250 °C and detector temperature 280 °C. Mass spectra were taken at 70 eV with a mass range of m/z 40–500. Identification of the components was achieved by comparison of the retention time and mass spectra of each separated peak with the databank of the instrument, NIST 2005 library and the literature.

2.5. Animals

Healthy male Wistar albino rats weighing between (100–150 g) and male Swiss albino mice (20–30 g) were used in this study. Animals were housed in polypropylene cages at controlled conditions of a light and dark cycle (12 h/12 h) and a temperature of 22 ± 2 °C. They were given pellet feed (Vital feed, UAC PLC, Nigeria) and water *ad libitum*. The experiments were performed after getting necessary approval from the Institutional Animal Ethical Committee (CM/COM/08/VOL.XXV) of University of Lagos and governed by the United States National Academy of Sciences Guide for the Care and Use of Laboratory Animals (2011).

2.6. Acute toxicity test

Acute toxicity of the ethanolic extract of *M. discoidea* was determined in male Swiss albino mice according to the method of Hilaly et al. (2004) with slight modifications. Mice fasted for 18 h were randomly divided into five groups of six mice per group. Graded doses of the extract (400, 800, 1600, 3200 and 5000 mg/kg) were separately administered by gavage using a suitable canula to the mice in each of the groups and the control group received distilled water (10 ml/kg). All animals were then observed for toxic symptoms and mortality for 24 h and then over a period of 7 days.

2.7. Gastroprotective activity

2.7.1. Effect of M. discoidea extract on ethanol-induced ulcer

The experiment was performed with slight modifications of the method reported by Kim et al. (2008). Rats were divided into five groups ($n=7$) and fasted for 24 h prior to oral administration of vehicle (water, 10 ml/kg), misoprostol (0.1 mg/kg) or extract (50, 100 and 150 mg/kg). One hour later, absolute ethanol (1 ml) was orally administered to the rats for the induction of gastric ulcer. Animals were sacrificed 1 h later after ethanol administration; their stomachs were removed and longitudinally excised along the greater curvature and rinsed thoroughly in normal saline. This was followed by macroscopic examination of the gastric mucosal for ulcer lesions. The number, length and severity of the ulcers were noted and scored on an arbitrary 0–6 point scale (Galati et al., 2001). The % of ulcer inhibition was calculated in relation to the ulcer index as follows: UI=Total ulcer score/No of ulcerated animals

% of inhibition $=(A_0-A_1)/A_0\times 100$ where A_0=ulcer index of control and A_1=ulcer index of treated group.

This model was also used for the screening of the fractions at the dose of 150 mg/kg. This dose gave the best activity in all the three models used in this study.

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2.7.2. Estimation of antioxidant parameters of stomach tissues in ethanol-induced ulcer

After the macroscopic analyses, malondialdehyde (MDA), reduced glutathione (GSH), superoxide dismutase (SOD) and catalase (CAT) enzyme activities in rat stomach tissues were determined. The stomach tissue homogenates (10%, w/v) were prepared in cold 50 mM potassium phosphate buffer (pH 7.4) and centrifuged using Thermo Scientific Heraeus Labofuge 200 Centrifuge at 4000 g for 10 min. Then, the supernatants were used for the determination of the enzymatic activities. All assays were carried out at room temperature and in triplicate.

Protein was estimated in mucosal homogenate for expressing the activities of MDA, GSH, SOD and CAT per mg of protein by the method of Lowry et al. (1951). Bovine serum albumin (BSA, 1 mg/ml) was used as a standard and measured in the range of 0.1–1 mg/ml.

Lipid peroxidation in the tissue homogenate was estimated by monitoring thiobarbituric acid reactive substance formation and expressed in terms of MDA equivalents using an extinction coefficient of 1.56×10^5/M/cm, according to the method of Ohkawa et al. (1979). The GSH content was determined as reported by Moron et al. (1979). Briefly, the homogenate was precipitated with 25% trichloroacetic acid (TCA) and centrifuged. The supernatant was used for GSH determination using freshly prepared 5,5-dithiobis (2-nitrobenzoic acid) (DTNB) solution. Superoxide dismutase (SOD) activity was estimated according to the method described by Misra and Fridovich (1972). For the determination of catalase, the decomposition of H_2O_2 in the presence of catalase was followed at 240 nm and the results expressed as unit/mg protein (Aebi, 1984).

2.7.3. Effect of M. discoidea extract on indomethacin-induced ulcer

The experiment was performed as described by Nwafor et al. (2000), with minor modifications. Briefly, male Wistar rats (120–150 g) were fasted for 48 h with free access to water before the start of the experiment. After 48 h of fasting, rats were treated with vehicle (distilled water, 10 ml/kg), omeprazole (100 mg/kg), or the extract (50, 100 and 150 mg/kg). An hour later, indomethacin (80 mg/kg, b.w.) was orally administered to induce maximum level of ulcer in all the groups. The animals were euthanized 6 h later after indomethacin treatment. Stomachs were removed, opened along the greater curvature, rinsed thoroughly in normal saline and severity of ulcer determined as described earlier.

2.7.4. Effect of M. discoidea on pylorus ligation-induced ulcer

Pylorus ligation was performed according to the method described by Shay et al. (1945). Male Wistar rats were deprived of food for 48 h before pyloric ligation, but had free access to water. After 1 h pre-treatment with the extract (50, 100 and 150 mg/kg), cimetidine (100 mg/kg) and vehicle (10 ml/kg), pyloric end of the stomach was ligated. Precaution was taken to avoid traction to the pylorus or damage to its blood supply. The stomach was placed carefully in the abdomen and the wound was sutured by interrupted sutures. After 4 h, the animals were sacrificed, their stomachs removed and longitudinally excised along the greater curvature. The inner surface was then examined for ulcerative lesions. The ulcer index and protective percentage was calculated.

To determine the gastric content, the entire gastric content was transferred into centrifuge tubes. The tubes were centrifuged at 1000 g for 10 min and the gastric volume was directly read from the graduation on the tubes. The supernatant was then collected and pH was determined using a digital pH meter. Total acidity was determined by titrating 1 ml of gastric juice against 0.1 N NaOH to pH 7 using phenolphthalein as the indicator and the concentration was expressed as mEq/l.

2.8. In vitro antioxidant capacity

Antioxidant activity of *M. discoidea* extract was evaluated using DPPH and hydroxyl radical scavenging assays (Mathew and Abraham, 2006; Luo et al., 2009). The ferrous ions chelating activity of the extract and standard (EDTA) were investigated according to the method of Dinis et al. (1994) while the method of Benzie and Strain (1996) was adopted for the ferric reducing antioxidant power assay. The extract was assayed at five different concentrations (10–50 μg/ml).

2.9. Statistical analysis

Statistical analysis was performed with Graph Pad Prism version 5.0 for Windows (GraphPad Software, San Diego, CA, USA) using one-way analysis of variance (ANOVA) followed by Tukey's Multiple Comparison Test. The significance levels were analyzed at $P < 0.01$, $P < 0.05$ and $P < 0.001$.

3. Results

3.1. Phytochemical study

Preliminary phytochemical tests showed the presence of saponins, tannins, alkaloids, glycosides, flavonoids and anthraquinones. The content of total phenols, flavonoids, and proanthocyanidins in the extract was 142.47 mg/g gallic acid equivalent, 8.03 mg/g quercetin equivalent and 171.77 mg/g catechin equivalent of the dried extract, respectively.

3.2. Chemical characterization of the active hexane fraction of M. discoidea by GC–MS

The chromatogram and chemical constituents of the hexane fraction are shown in Fig. 1 and Table 1. A total of 14 compounds were detected in the extract. Ethyl esters of palmitic, linoleic and stearic acids were detected in relative concentrations of 8.96, 9.38 and 4.40% of the total fraction, respectively. The extract also showed GC peaks of oleic acid and 1,2-Benzenedicarboxylic acid, mono (2-ethylhexyl) ester at 32.18 and 38.62 min, respectively.

3.3. Acute toxicity

Oral administration of the extract to mice at different doses did not show any signs of toxicity and mortality was not recorded even at the highest dose of 5000 mg/kg.

3.4. Effect of M. discoidea on acute gastric mucosal lesions induced by ethanol

The effect of orally administered ethanolic extract of *M. discoidea* on gastric damage induced by absolute ethanol is shown in Table.Oral administration of absolute ethanol to the control animals produced multiple mucosal lesions in the rat stomach in the form of hemorrhagic streaks with ulcer index of 3.5. Pretreatment with the extract at the dose of 50, 100 and 150 mg/kg showed 46.94%, 68.51% and 76.68% protection compared to the control group, respectively. Misoprostol also offered significant ($P < 0.05$) protection (94.75%, inhibition) against ethanol induced gastric lesions and the effect was higher than that of the extract. In this model, pretreatment with the solvent fractions at the dose of 150 mg/kg showed hexane fraction to be the most potent, with ulcer inhibition of 70.59%, compared to the control (Table 2).

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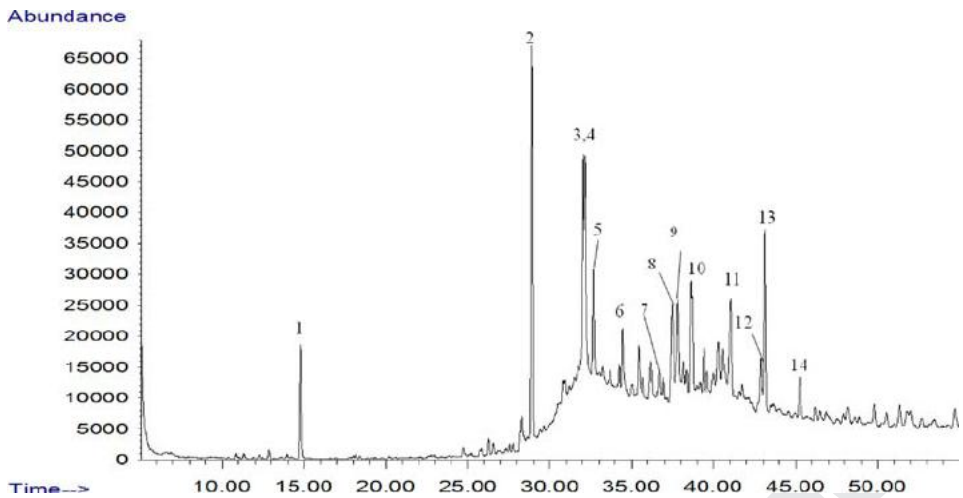

Fig. 1. Gas chromatogram of the hexane fraction of *M. discoidea*.

Table 1
Chemical composition of the hexane fraction of *M.discoidea*.

	RT	Name of the compound	MW	Area %	Fragment ions
1	14.81	Not identified	156	5.36	**121**,156,91,77,158,51,65,50,63,53
2	28.89	Palmitic, ethyl ester	284	8.96	**88**,101,55, 57,69,73, 70, 89,83,61
3	32.06	Linoleic, ethyl ester	308	9.38	**67**, 55, 81,79,69, 95,54,82,68, 96
4	32.18	Oleic acid	282	3.82	**55**, 69,67,83,79, 81, 97,57,95, 88
5	32.68	Stearic acid, ethyl ester	312	4.40	**88**,55,101,57,69,73,83,60,70,71
6	34.46	5-(Prop-2-enoyloxy)pentadecane	282	1.33	**55**,41,43,57, 39,72,56,44,40,42
7	37.53	Thiophene, 3-(methylthio)-	130	1.23	**130,** 45,115,71,97,44,84,39,69,85
8	37.83	Trispiro[(3,4-diazatetracyclo [5.5.1.0(2,6)0.0(8,12)] trideca-3,10-diene)-5,9,13-tricyclopropane]	252	2.63	**117,** 91,132,131,39,115,77,105,104,128
9	38.62	1,2-Benzenedicarboxylic acid, mono (2-ethylhexyl) ester	278	1.51	**149,** 67,57,71,43,70,41,150,55,113
10	38.70	Not identified	300	3.82	**185**,69,213,55,300,57,128,115,129,187
11	41.03	Ditetradecyl ether	410	2.11	**57,** 71, 43,83, 85, 197, 85,69,55, 97, 82
12	42.87	3-Acetoxytridecane	242	1.11	**43**, 41,55,57,101,69, 56,83,97,71
13	43.09	Not indentified	312	9.09	**225**,253,239,240,165,152,268,210,226,55
14	45.28	Methoxyacetic acid, tridecyl ester	272	0.83	**45, 57,** 43, 71,41,55, 29, 85,69, 27

The *m*/*z* column reports the most abundant fragments observed in the mass spectrum (the base peaks are highlighted in bold).

3.5. Effect of M. discoidea extract on antioxidant enzyme activities

The effect of the extract on antioxidant enzyme activities is shown in Table 3. The control group had the highest MDA content with a value of 3.6 ± 0.15 nmol/mg protein, while the extract (50, 100 and 150 mg/kg) significantly ($P < 0.05$) decreased the MDA content to 2.40 ± 0.19, 1.80 ± 0.16 and 1.25 ± 0.03 nmol/mg protein, respectively. The activity of all antioxidant enzymes (SOD, GSH and CAT) decreased in the gastric mucosa of animals exposed to ethanol (the control group). The reduction was significantly ($P < 0.05$) improved by the administration of the extract, compared to the control.

3.6. Effect of M. discoidea extract on indomethacin-induced ulcer

The extract at the dose of 50, 100 and 150 mg/kg prevented the acute gastric ulcers in a dose dependent manner (Fig. 2). The extract exerted 65.41%, 81.13% and 95.28% gastroprotective effect, respectively. Omeprazole significantly inhibited the incidence and severity of ulcer by 96.23%.

3.7. Effect of M. discoidea extract on pylorus ligation-induced ulcer

In this model, only the dose of 150 mg/kg of the extract produced significant decreased in ulcer index and an increase in pH of the gastric juice compared to the control (Table 4). The treatment with the extract did not significantly reduce gastric juice volume and total acidity. On the other hand, reference drug, cimetidine significantly reduced ulcer index, gastric volume, total acidity and increase gastric pH compared to the control.

Table 2
Effect of ethanolic stem bark extract and fractions of *M.discoidea* in ethanol-induced gastric ulcer in rats.

Treatment	Dose (mg/kg)	Mean ulcer score	Ulcer index	% Inhibition
Control	10 ml/kg	3.5 ± 0.82	3.50	–
M. discoidea	50	1.86 ± 0.46	1.86	46.94
	100	1.29 ± 0.47*	1.10	68.51
	150	1.14 ± 0.54*	0.82	76.68
Misoprostol	0.1	0.43 ± 0.20**	0.18	94.75
Fractions				
Control	10 ml/kg	5.67 ± 0.56	5.67	-
HEX	150	1.67 ± 0.47*	1.67	70.59
EtOAc	150	4.50 ± 0.72	4.57	19.33
DCM	150	5.42 ± 0.66	5.42	4.41
BuOH	150	2.83 ± 1.00	2.36	58.33
AQU	150	5.17 ± 1.22	5.17	8.82

Results are mean ± SEM, extract ($n = 7$); fractions ($n = 6$). Statistical comparison was determined by one way ANOVA followed by Tukey's multiple comparison tests.

* $P < 0.05$ statistically significant compared to the control.
** $P < 0.01$ statistically significant compared to the control.

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Table 3
Effect of ethanolic stem bark extract of *M. discoidea* on antioxidant enzyme activities in ethanol-induced gastric ulcer in rats.

Treatment	Dose (mg/kg)	SOD (U/mg protein)	GSH (nmol/mg protein)	CAT(U/mg protein)	MDA (nmol/mg protein)
Control	10 ml/kg	8.29 ± 0.66	18.21 ± 0.22	2.67 ± 0.12	3.60 ± 0.15
M. discoidea	50	12.52 ± 1.04***	20.73 ± 0.38	2.88 ± 0.16	2.40 ± 0.19***
	100	14.0 ± 0.44***	21.50 ± 0.46*	4.19 ± 0.16	1.80 ± 0.16***
	150	15.34 ± 0.59***	24.80 ± 1.05***	5.06 ± 0.05***	1.25 ± 0.03***
Misoprostol	0.1	17.21 ± 0.56***	25.61 ± 1.11***	5.20 ± 0.11***	1.13 ± 0.08***

Values are expressed as mean ± SEM ($n=6$). Statistical comparison was determined by one way ANOVA followed by Dunnett's multiple comparison tests.
* $P<0.05$ statistically significant compared to the control.
*** $P<0.001$ statistically significant compared to the control.

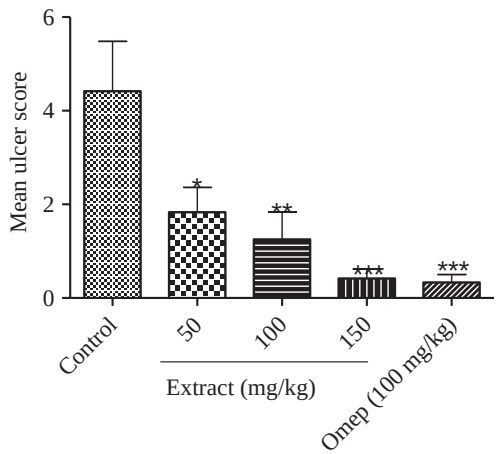

Fig. 2. Effect of ethanolic stem bark extract of *M. discoidea* and Omeprazole (omep) in indomethacin-induced ulcer model. Results are mean ± SEM ($n=6$). *$P<0.05$, **$P<0.01$, ***$P<0.001$ *versus* control. One-way ANOVA followed by Dunnett's test.

3.8. *In vitro antioxidant assays*

The antioxidant activity of the extract monitored *in vitro* by the DPPH and hydroxyl radical scavenging, ferric reducing antioxidant power (FRAP) and ion-chelating assays is presented in Fig. 3. The scavenging effect of the extract on DPPH and hydroxyl radicals was measured as 77.40% and 70.80% at 50 μg/ml, respectively. The % chelation produced by the extract were 17.79%, 36.71%, 46.00%, 58.22% and 71.46% at a concentration of 10, 20, 30, 40 and 50 μg/ml, respectively. The FRAP values were 101.6 ± 27.7 and 136.2 ± 14 for the extract and quercetin, respectively.

4. Discussion and conclusion

Gastric ulcer disease is a multi-factorial disease and the significant role played by reactive oxygen species and free radicals during its pathogenesis is well experimented in both human and experimental animals (Nartey et al., 2012). In the present study, we investigated the gastroprotective activity of *M. discoidea* in ethanol, indomethacin and pylorus ligation models. The antioxidant potential of the extract was evaluated *in vivo* in ethanol-induced model and in *in vitro* assays. The components present in the active hexane fraction from the crude extract of *M. discoidea* were analyzed by GC–MS system.

The acute toxicity test in mice of *M. discoidea* extract did not showed observable toxic effect and mortality was not recorded even at 5000 mg/kg. This finding correlates with the report of Adedapo et al. (2010), indicating that the extract could be regarded as safe at the doses tested. Moreover, according to OECD (1981) guideline and Hayes (1989), no dose-related toxicity by oral route should be considered above 5 g/kg body weight

Administration of absolute ethanol by gavage has long been used as a reproducible method to induce gastric injury in experimental animals (Yu et al., 2014). Ethanol as an ulcerogen causes severe gastric mucosal ulceration either by acting directing on the gastric mucosal, altering protective factors or indirectly by increasing the release of vasoactive products such as histamine from mast cells (Szabo, 1987; Goulart et al., 2005; Yu et al., 2014). In this study, the exposure of the gastric mucosa to ethanol led to the development of widespread acute gastric lesions while pretreatment with different doses of *M. discoidea* significantly attenuate the gastric ulceration. The result indicates a possible cytoprotective activity. The concept of gastric cytoprotection signifies protection against mucosal injury by a mechanism other than inhibition of acid secretion (Robert, 1979; Sashidhara et al., 2014).

Ethanol causes severe oxidative stress in gastric tissue and potent antioxidant defense is absolutely essential for gastric protection (La Casa et al., 2000). The endogenous anti-oxidant enzymes (SOD, CAT and GSH) in the gastric mucosa are the key components of cellular defense system against reactive oxygen species (ROS) (Nartey et al., 2012). SOD is the cell's first line of defense against ROS. It acts by transforming ROS into less reactive hydrogen peroxide and protects the cells from damage through the removal of the excess radicals. The SOD activity level represents the intracellular antioxidant ability of the cell (Bonamin et al., 2011). Catalase controls the accumulation of ROS generated through numerous metabolic processes such as the breakdown of ethanol in the stomach. Inhibition of catalase activity leads to lipid formation by increasing the generation of hydroxyl radicals (Lee et al., 2012). The biochemical function of GSH is to reduce lipid hydroperoxides to their corresponding alcohols and free hydrogen peroxide to water, hence plays an important role in cell protection (Nguyen et al., 2013). Our results showed that the extract increased the level of SOD, CAT and GSH activities and reduced lipid peroxidation. These results indicate that the extract may exert its gastroprotective effect *via* an antioxidant mechanism.

In-vitro antioxidant tests using different assays also showed that the extract has good free radical scavenging potential. The results provide considerable support to the antioxidative potential of the extract. It is therefore possible that the gastroprotective effect observed with this extract may be attributed to the potent anti-oxidant activity.

The indomethacin-induced ulcer model represents a form of gastric irritation resulting from the inhibition of prostaglandin synthase (Steinmeyer, 2000). The inhibition of prostaglandin synthesis is known to be the main ulcerogenic mechanism of non-steroidal anti-inflammatory drugs (NSAID), besides provoking damage to the vascular endothelium, reduction of the blood flow, formation of obstructive micro-thrombi and activation of neutrophils (Guth, 1992). On the other hand, increase in prostaglandins especially prostaglandin E_2 and prostaglandin I_2 has been associated with cytoprotection (Deshpande et al., 2003). Therefore, agents that inhibit the effects of NSAID would exhibit

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Table 4
Effect of ethanolic extract of *M. discoidea* on pylorus ligation-induced gastric ulcer in rats.

Treatment	Dose (mg/kg)	Mean ulcer score	Ulcer index	% Inhibition	Gastric volume (ml)	Gastric pH	Gastric acidity (meq/l)
Control	10 ml/kg	1.10 ± 0.10	1.10	–	0.84 ± 0.05	4.12 ± 0.19	7.80 ± 0.04
Extract	50	0.60 ± 0.10	0.60	45.46	0.62 ± 0.10	4.24 ± 0.07	7.80 ± 0.04
	100	0.70 ± 0.30	0.42	61.82	0.88 ± 0.17	4.10 ± 0.08	8.40 ± 0.010
	150	0.20 ± 0.12**	0.08	92.73	0.52 ± 0.19	4.80 ± 0.29*	5.40 ± 0.09
Cimetidine	100	0.10 ± 0.10**	0.02	98.18	0.22 ± 0.13*	7.48 ± 0.12***	4.00 ± 0.06**

Values are expressed as mean ± SEM ($n=5$).
* $P<0.05$ when compared with control group (one way ANOVA, followed by Dunnett's multiple comparisons test).
** $P<0.01$ when compared with control group (one way ANOVA, followed by Dunnett's multiple comparisons test).
*** $P<0.001$ when compared with control group (one way ANOVA, followed by Dunnett's multiple comparisons test).

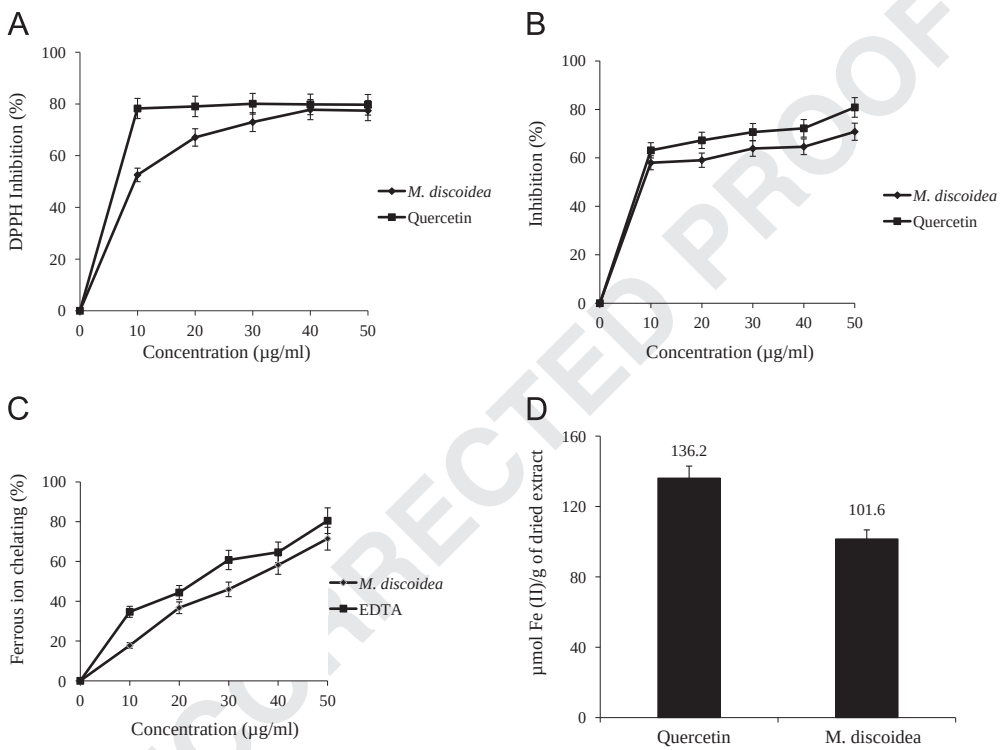

Fig. 3. *In vitro* antioxidant activities of ethanolic stem bark extract of *M. discoidea*. A-DPPH radical scavenging, B-hydroxyl radical scavenging, C-ferrous ion chelating, D-ferric reducing antioxidant power (FRAP).

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cytoprotection. The results obtained in this model showed that the extract significantly inhibits the gastric effect of indomethacin in a dose dependent manner, which might also contribute favorably to its gastroprotective property.

In pylorus ligation-induced model, gastric acid is an important factor responsible for the production of ulceration (Zhang et al., 2014). It has been shown that the volume of gastric acid produced determines the permeability of the gastric mucosal wall which triggers and accelerates the development of ulcers (Khan et al., 2014). Pylorus ligation also produces mucosal damage by altering the level of cytoprotective prostaglandins and cytokines as well as interfering with the gastric mucosal resistance (Singh et al., 2008). Inhibition of acid production maintains the permeability of the gastric mucosal membrane and help to reduce the formation of gastric ulcers. Pre-treatment with the ethanolic extract of *M. discoidea* only showed significant ulcer protection and increase in gastric pH at the dose of 150 mg/kg. There was no significant reduction in the gastric juice volume and total acidity. This observation largely shows that the extract does not act in the mechanisms that control the production of gastric acid. The results therefore, suggest that *M. discoidea* did not protect the gastric mucosa through anti-secretory action.

The screening of the solvent fractions using ethanol-induced model indicated hexane fraction to be the most active, hence GC–MS analysis was done to characterize the chemical composition of this non-polar fraction. Ethyl esters of palmitic and linoleic acids were found as the major compounds in this fraction. Recently, linoleic acid was reported as one of the major compounds in the cytotoxic active ethyl acetate fraction from this plant (Johnson-Ajinwo et al., 2015). It is well known that various fatty acids have anti-inflammatory and antioxidant activities. Palmitic acid, ethyl ester has been reported to possess antioxidant and anti-inflammatory activities (Roy et al., 2011). A recent work published by Martins et al. (2014) suggests the involvement of anti-inflammatory hydroxyl-linolenic, linoleic acids and conjugated oxo-linoleic acids in the observed inhibitory effect of the hexane fraction of *Celtis iguanaea* in different models of induced ulcers. Therefore, the observed activity of the fraction could be attributed to the presence of these fatty acids. However, the presented data cannot confirm that the gastroprotective activity of *M. discoidea* is attributable to the single or synergistic action of these components. Further studies

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are required to find out the active compounds responsible for the observed gastroprotective activity and expand the mechanism of action.

In conclusion, the study revealed that ethanolic extract of *M. discoidea* possess gastroprotective potential, which could be mediated through antioxidant mechanism. The results of this study provide scientific support for the use of this species for the treatment of gastric ulcer.

Q3 Uncited references

(Committee for the Update of the Guide for the Care and Use of Laboratory Animals, 2011).

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